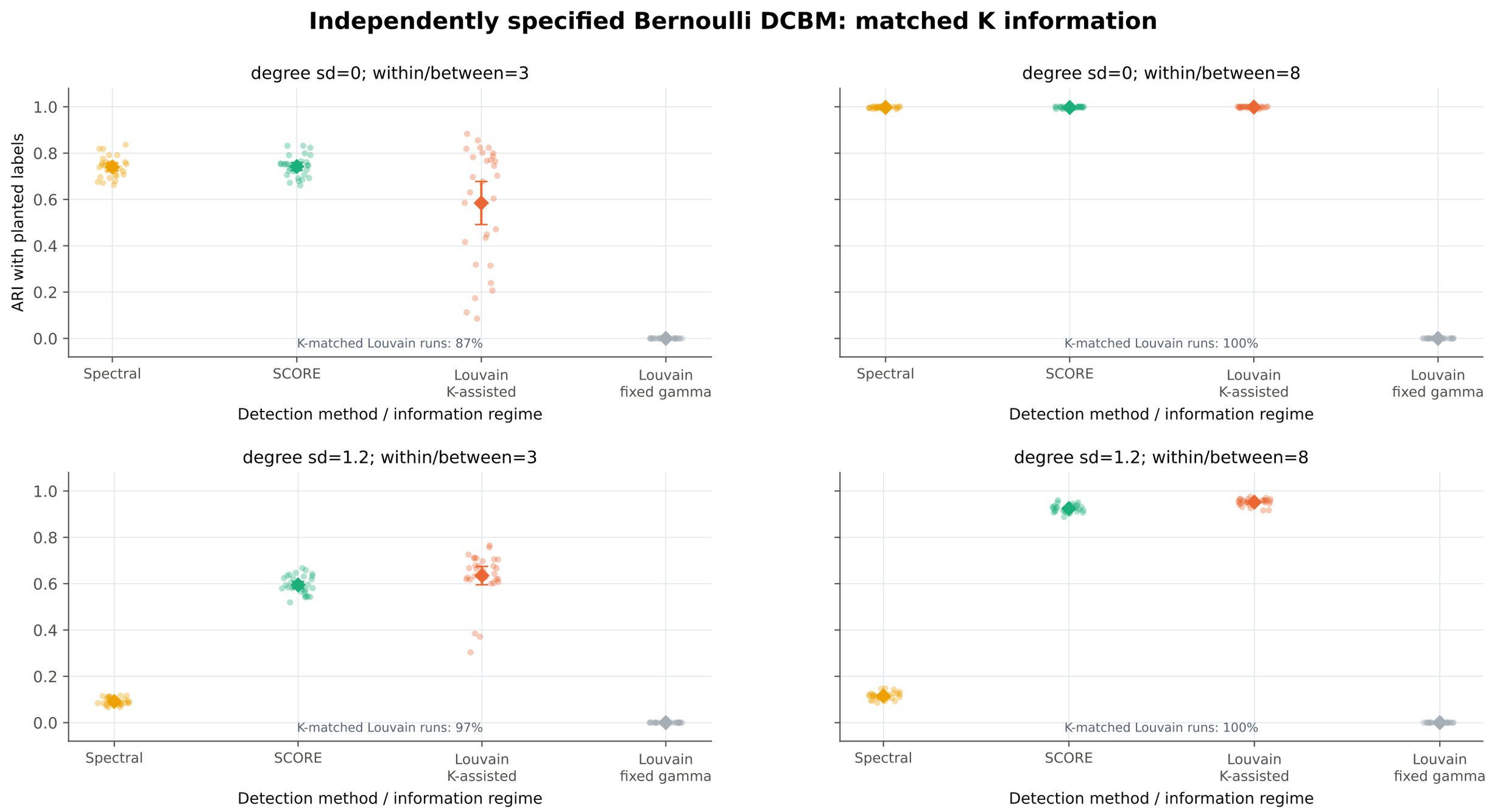
Independently specified Bernoulli DCBM: matched K information
degree sd=0; within/between=3
degree sd=0; within/between=8
degree sd=1.2; within/between=3
degree sd=1.2; within/between=8
ARI with planted labels
1.0
0.8
0.6
0.4
0.2
0.0
K-matched Louvain runs: 87%
K-matched Louvain runs: 100%
K-matched Louvain runs: 97%
K-matched Louvain runs: 100%
Spectral
SCORE
Louvain
K-assisted
Louvain
fixed gamma
Detection method / information regime
30 paired graph replications per setting; three planted groups; expected full-graph mean degree 18.
Same connected evaluation population across degree/signal regimes within each replication; probabilities are never clipped and graphs are not redrawn.
Diamonds/bars: mean and pointwise Monte Carlo 95% interval. Gray fixed-resolution Louvain is a reference, not part of the matched-information comparison.